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The future might resemble the past

With mounting concern over the depleting rain forest cover in the equatorial regions, a significant number of researchers, corporate entities and scientists all over the world, have been working towards developing a viable alternative to conventional stationery.

In the early 70s, Nicholas K Sheridan, then a researcher at the Xerox Palo Alto Research Center, envisaged flexible computer screens made out of flexible materials such as paper. His contention was that bendable screens would have sharper, clearer displays, and consume less power. The idea stuck.

At Gyricon's Ann Arbor, Michigan plant, Engineer Mark Edmond is working on a 'display image board'. The product under development by this Xerox subsidiary is SmartPaper—thin and flexible polymer sheeting that can behave precisely like a computer screen.

On April 8, 2004, IXYS Corporation announced that its Micronix division had unveiled the industry's first standard product driver for electronic paper displays based on E Ink Corporation's electronic ink technology. The new display driver IC, the MX834, is targeted at the emerging market of displays that are easy to read and consume less power at the same time.

The commercialisation of electronic paper technology is the result of a collaboration started in 2001 between Elnk Corporation, Toppan Printing and Philips, together with Sony. Elnk uses a process similar to Gyricon's, and has been focusing on the consumer electronics market.

Philips has come up with something that uses a form of electronic ink devised in 1998 by a team at MIT. Their e-ink consists of hollow, transparent plastic capsules containing black and white pigment powders. Accordingly, they set up a company, Polymer Vision, to turn their approach into an industrial process. Polymer Vision is now setting up a pilot production line, and hopes to be

soon producing more than 5,000 e-paper displays a year.

Sales of flexible screens would be about US\$ 18 million by 2010 as opposed to US\$ 311,000 last year, a 60-fold increase, according to California-based iSuppli Corporation. However, this pales in comparison to the 29 per cent compound annual growth rate (CAGR) of LCD revenues to over US\$26 billion by the end of 2005, according to DisplaySearch, the FPD market research and consulting firm.

Market turnstiles: Whose tables will turn?

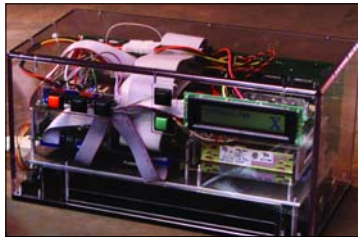
According to an American survey, the most promising market for these new-age products is the sign-painting market. E-paper could also enable dailies to save on publishing expenses. Digital newspapers may never get to broadsheet or even tabloid size, reports Newsweek, the magazine: "Costs would be too high. Initially, digital newspapers would merely receive data from a PC or a wireless hotspot."

On a different note, Elnk Corp introduced an electronic book reader with Sony in March, the Sony Librie. This has the ability of reflecting natural light—the most important characteristic of paper—and its surface texture has a semblance to newsprint.

Innovative use of electronic paper and ink will help ease the predicament of the publishing industry, now thoroughly dependent on pulp paper, and dangerously at loggerheads with powerful environment groups.

Companies such as Xerox, Eastman Kodak Co, Royal Philips Electronics, Sony Corp, etc, have already launched ventures to get their share of the e-paper pie. Researchers from Stanford University and MIT have also expressed interest.

As Bruce Rutledge, President, Chin Music Press, Seattle, suggests, the simple pleasures of life will still retain their appeal—it would be quite some time before electronic books unseated printed ones from libraries. Bookworms need not begin digging burrows as yet.



PEOPLE TO WATCH

The Affordable Solutions evangelist and IIT-Bombay KReSIT (Kanwal Rekhi School of Information Technology) Head Deepak B Phatak, is spearheading a move to get viable computing solutions for Indians.

Computing is a big part of this and Phatak has just started on his dream project—that of getting final year engineering students to work on developing useful, quality open source applications.

However, open source isn't everything in such solutions: "I'd work with Microsoft if it means a more affordable solution", says Phatak, who was appointed to the prestigious Subrao Nilekani Chair in the KReSIT School. Here's to more of computing for the common man.

Prof. Deepak B Phatak,
HEAD PROFESSOR, KReSIT



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Update: In the 'People to Watch' box on Page 52, Krishna Bharat's designation has been given as 'Head, Google India. Unfortunately, Google clarified after the page went to print that Krishna Bharat is, in fact, one of the leaders of a team that is setting up the operation in Bangalore. His official designation is Principal Scientist, Google.